

Phonetics

Introduction to linguistics — **Course Website**

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Outline

1. **Phonetics vs phonology**: What's the difference between sounds we make and sound systems?
2. **Speech organs**: Which body parts do we use to produce speech?
3. **Consonants**: How do we classify consonant sounds?
4. **Vowels**: How do we describe and map vowel sounds?
5. **Transcription**: How can we write down pronunciation?

Phonetics vs phonology

phonetics	phonology
sounds as such language use (<i>parole</i>) not language-specific substance concrete phone []	sounds as parts of a sound system language system (<i>langue</i>) language-specific function (meaning differentiation) abstract phoneme / /

Kortmann (2020): 28

Exercise

Assign the following items to the correct linguistic concepts:

- langue
- slashes for transcription /.../
- not language-specific
- concrete realisation
- abstract entity
- parole
- square brackets for transcription [...]
- language-specific

Concepts

1. **phoneme**

2. **phone**

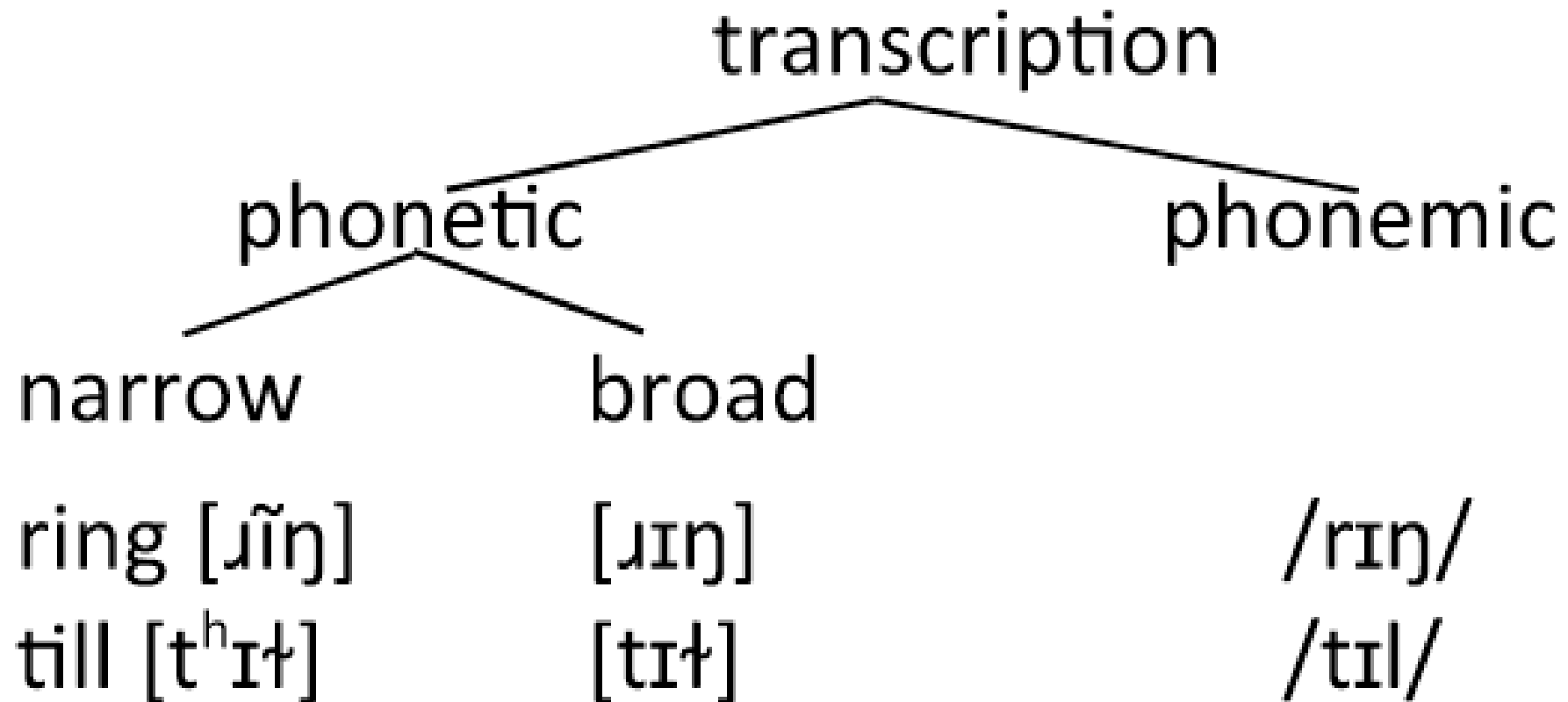
Phoneme:

- langue
- language-specific
- slashes for transcription /.../
- abstract entity

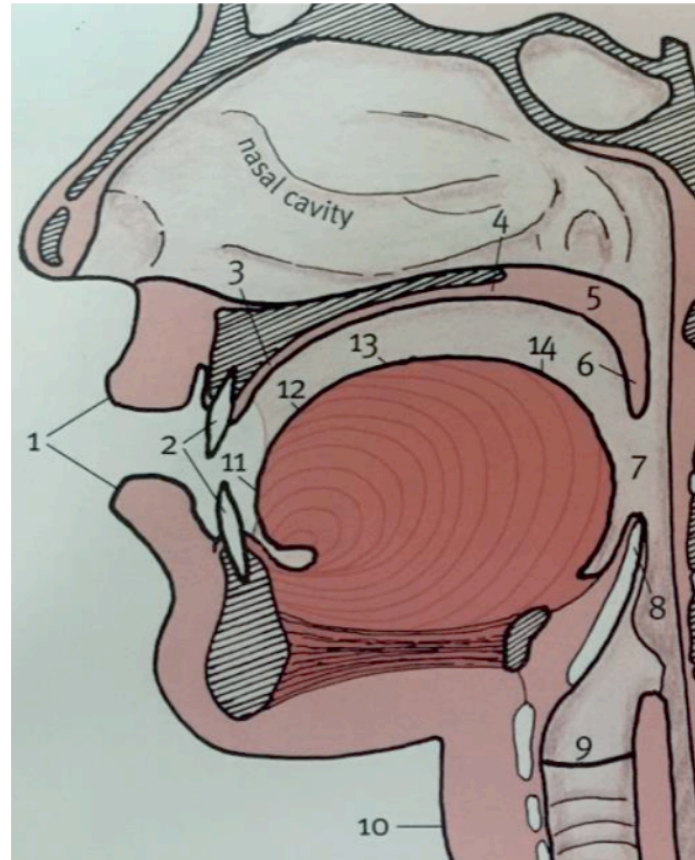
Phone:

- concrete realisation
- not language-specific
- square brackets for transcription [...]
- parole

Broad vs narrow transcription



Speech organs

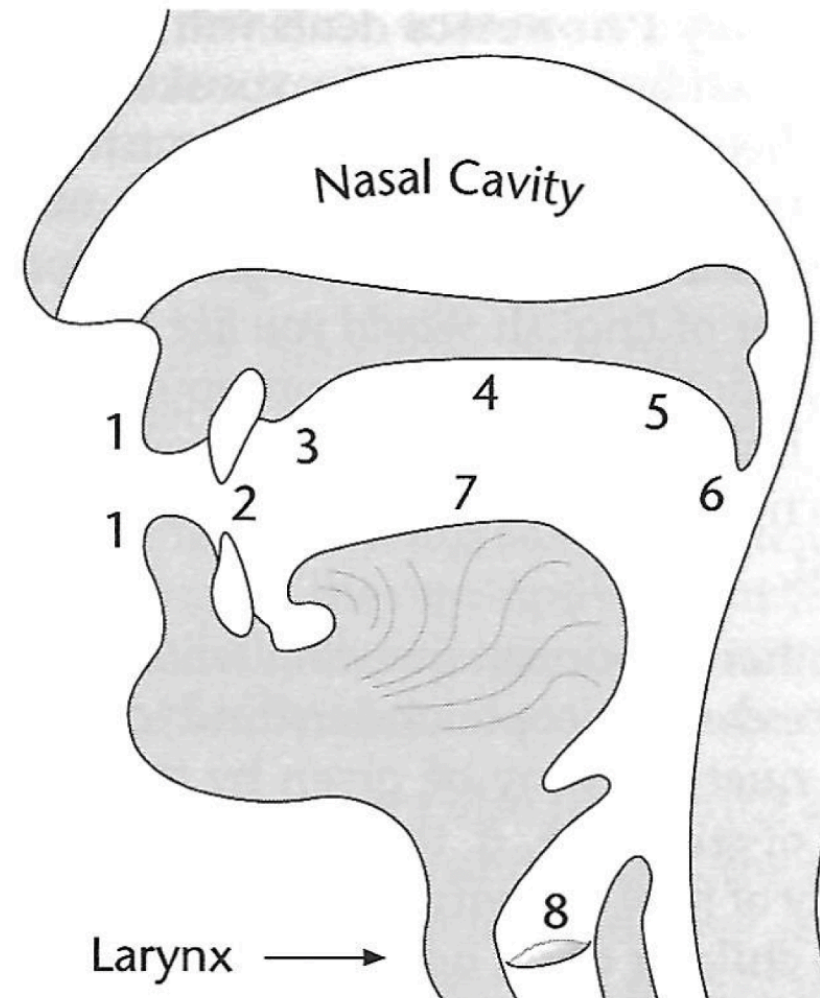


Speech organs

1. lips
2. teeth
3. alveolar ridge
4. hard palate
5. soft palate (velum)
6. uvula
7. pharynx
8. epiglottis
9. glottis – gap between vocal folds
10. larynx
11. tip/apex of the tongue
12. blade/lamina of the tongue
13. front of the tongue
14. back/dorsum of the tongue

Exercise: name vocal organs and describe speech sounds

1. Fill in the blanks in this figure with the correct names of the vocal organs.
2. Give the respective adjectives used to describe speech sounds produced at this place.



No.	Organ	Adjective
1	lips	labial
2	teeth	dental
3	alveolar ridge	alveolar
4	hard palate	palatal
5	soft palate (velum)	velar
6	uvula	uvular
7	tongue	dorsal
8	glottis	glottal

Place of articulation

	place of articulation		adjective	examples of sounds []	examples
1	lips		labial	[p], [b], [m], [w]	puppy, Bob, mummy, word
2	teeth		dental labio-dental	[θ], [ð] [f], [v]	throne, the fan, van
3	alveolar ridge		alveolar post-alveolar palato-alveolar	[t], [d], [s] [z], [n], [l] [ɹ] [ʃ], [ʒ], [tʃ], [dʒ]	turtle, dragon, sister zeal, nasty, lollipop rap shanty, illu <u>s</u> ion, chips, gipsy
4	hard palate		palatal	[j]	yellow
5	soft palate / velum		velar	[k], [g], [ŋ]	ketchup, gorilla, <u>s</u> ong
6	uvula		uvular		
7	pharynx		pharyngeal		
8	epiglottis				
9	glottis		glottal	[h], [ʔ]	hat, <i>in many non-RP accents</i> : bottle [bɒʔl]
10	larynx		laryngeal	[ʔ]	<i>sometimes in RP before /p, t, k/</i> : popcorn ['pɒʔp-kɔ:n]
11	tongue	tip/apex	apical	[θ], [ð], [ɹ], [l]	throne, the, rap, lollipop
12		blade/lamina	laminal	[t], [d], [n] [s], [z] [ʃ], [ʒ], [tʃ], [dʒ]	turtle, dragon, nasty sister, zeal shanty, illu <u>s</u> ion, chips, gipsy
13		front	(medio-)dorsal	[j], [ɪ], [i:], [e], [æ]	yellow, insect, <u>l</u> eap, pet, <u>p</u> at
14		back/dorsum	(post-)dorsal	[k], [g], [ŋ] [ʊ], [u:], [ɒ], [ɔ:]	ketchup, gorilla, <u>s</u> ong <u>f</u> oot, <u>f</u> ood, <u>t</u> on, <u>t</u> all

Kortmann (2020): 31

Consonants

Consonant inventory

place of articulation			bilabial	labio-dental	dental	alveolar	post-alveolar	palato-alveolar	palatal	velar	glottal
manner of articulation			vl. vd.	vl. vd.	vl. vd.	vl. vd.	vl. vd.	vl. vd.	vl. vd.	vl. vd.	vl. vd.
OBSTRU- ENTS		plosive	p b			t d				k g	
		fricative		f v	θ ð	s z		ʃ ʒ			h
		affricate						tʃ dʒ			
SONORANTS	APPROXIMANTS	LIQUIDS	nasal	m			n			ŋ	
			lateral				l				
			roll					r RP [ɹ] GA [ɹ]			
			semi-vowel	w					j		

Kortmann (2020): 33

Voiced vs unvoiced



a



b

Figure 2.2
a: voiced,
b: voiceless

Exercise: identify consonant types

Identify the IPA symbols below which represent

a. plosives

b. fricatives

c. voiced consonants.

1. [b]

2. [s]

3. [ʊ]

4. [w]

5. [ʃ]

6. [x]

7. [k]

8. [l]

9. [ɪ]

10. [θ]

11. [ŋ]

12. [d]

a. Plosives

- 1. [b]
- 7. [k]
- 12. [d]

b. Fricatives

- 2. [s]
- 5. [ʃ]
- 6. [x]
- 10. [θ]

c. Voiced consonants

- 1. [b]
- 4. [w]
- 8. [l]
- 11. [ŋ]
- 12. [d]

Other

- 3. [ʊ] (vowel)
- 9. [ɪ] (vowel)

Exercise: consonant inventory (Interactive)

Max challenge attempts exceeded. Please refresh the page to try again!

Exercise: consonant inventory (Reference)

Which of these ...

1. starts with an approximant: *you* or *do*?
2. ends with a fricative consonant: *race* or *real*?
3. medial consonant is voiced: *massive* or *leisure*?
4. starts with a dental consonant: *shy* or *thigh*?
5. starts with a bilabial consonant: *set* or *bet*?
6. ends with a stop consonant: *lip* or *pill*?
7. ends with a nasal consonant: *dumb* or *dull*?
8. starts with an alveolar consonant: *quip* or *dip*?
9. medial consonant is voiced: *robber* or *hopper*?
10. starts with a velar consonant: *lot* or *cot*?
11. starts with a labiodental consonant: *mat* or *vat*?

Word	IPA	Type
<i>you</i>	/j/	approximant
<i>race</i>	/s/	fricative
<i>leisure</i>	/ʒ/	voiced
<i>thigh</i>	/θ/	dental
<i>bet</i>	/b/	bilabial
<i>lip</i>	/p/	stop
<i>dumb</i>	/m/	nasal
<i>dip</i>	/d/	alveolar
<i>robber</i>	/b/	voiced
<i>cot</i>	/k/	velar
<i>vat</i>	/v/	labiodental

Exercise: determine consonant features

Determine the medial consonant phone(s) in each of the following words in terms of

- a. place of articulation
- b. manner of articulation
- c. voicing

1. *etching*
2. *father*
3. *dagger*

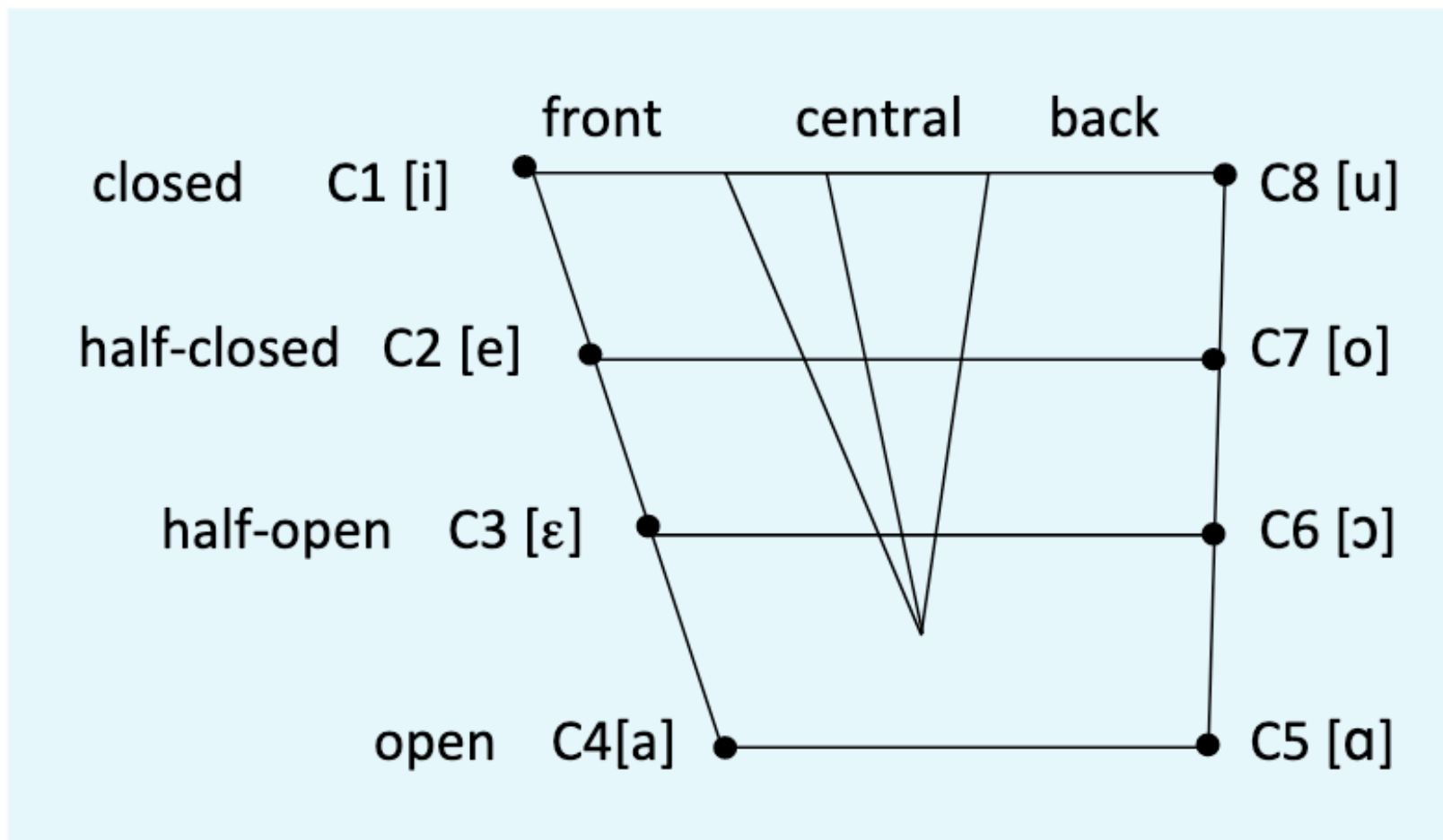
4. *lodger*
5. *sunny*
6. *hopper*

7. *pleasure*
8. *selling*
9. *singer*

Word	Symbol(s)	Place	Manner	Voicing
<i>etching</i>	tʃ	palato-alveolar	affricate	voiceless
<i>father</i>	ð	dental	fricative	voiced
<i>dagger</i>	g	velar	plosive	voiced
<i>lodger</i>	dʒ	palato-alveolar	affricate	voiced
<i>sunny</i>	n	alveolar	nasal	voiced
<i>hopper</i>	p	bilabial	plosive	voiceless
<i>pleasure</i>	ʒ	palato-alveolar	fricative	voiced
<i>selling</i>	l	alveolar	lateral	voiced
<i>singer</i>	ŋ	velar	nasal	voiced

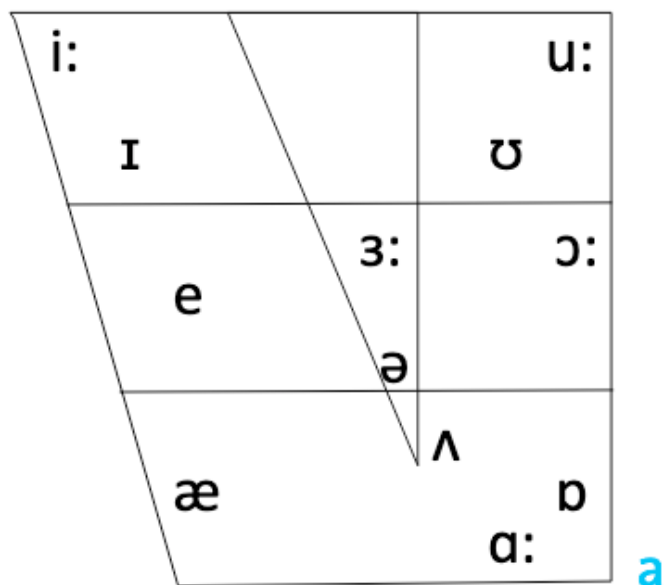
Vowels

Vowel chart: features

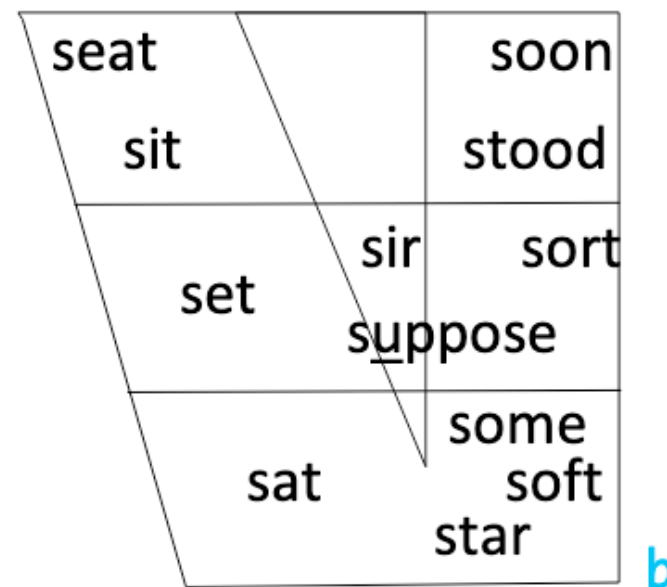


Vowel chart: English vowels & examples

Kortmann (2020): 36



Kortmann (2020): 36



Vowel differences between British and American English

Example	British English (RP)	American English (GA)
<i>pot</i>	/ɒ/	/ɑː/
<i>autumn</i>	/ɔː/	(sometimes) /ɑː/
<i>bird</i>	/ɜː/	/ɜr/
<i>mother</i>	/-ə/	/-ər/
<i>peer, pair, poor</i>	/ɪə, eə, ʊə/	/ɪr, er, ʊr/
<i>home</i>	/əʊ/	/oʊ/
<i>pass, ask, dance</i>	/ɑː/	/æ/
<i>fault</i>	/ɔː/ (closed)	/ɔr/ (more open)

Diphthongs

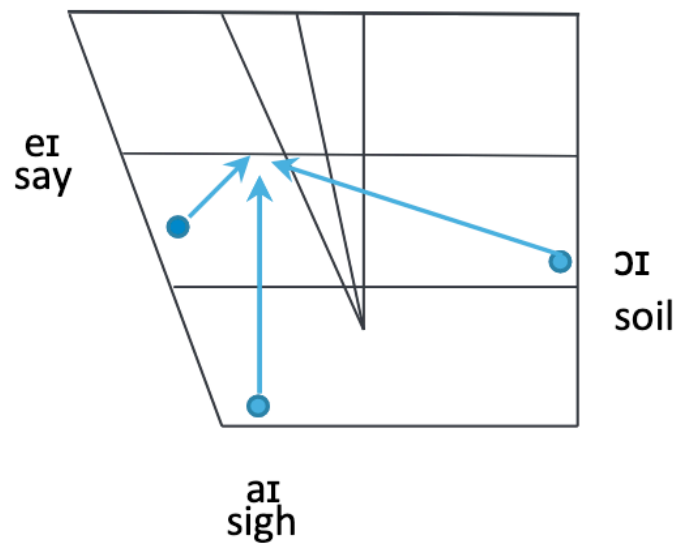


Figure 2.7: Closing diphthongs

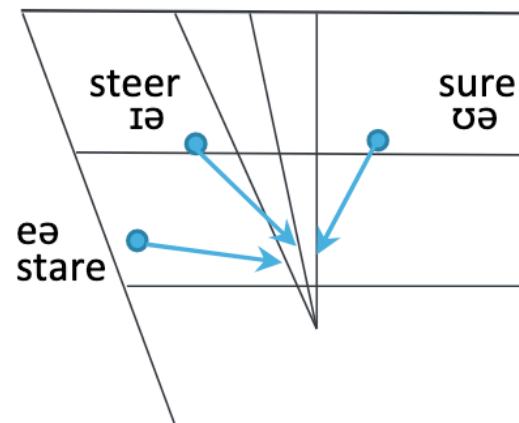


Figure 2.8: Centring diphthongs

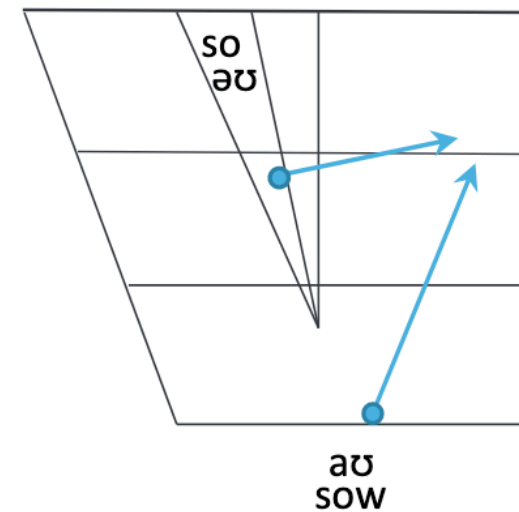
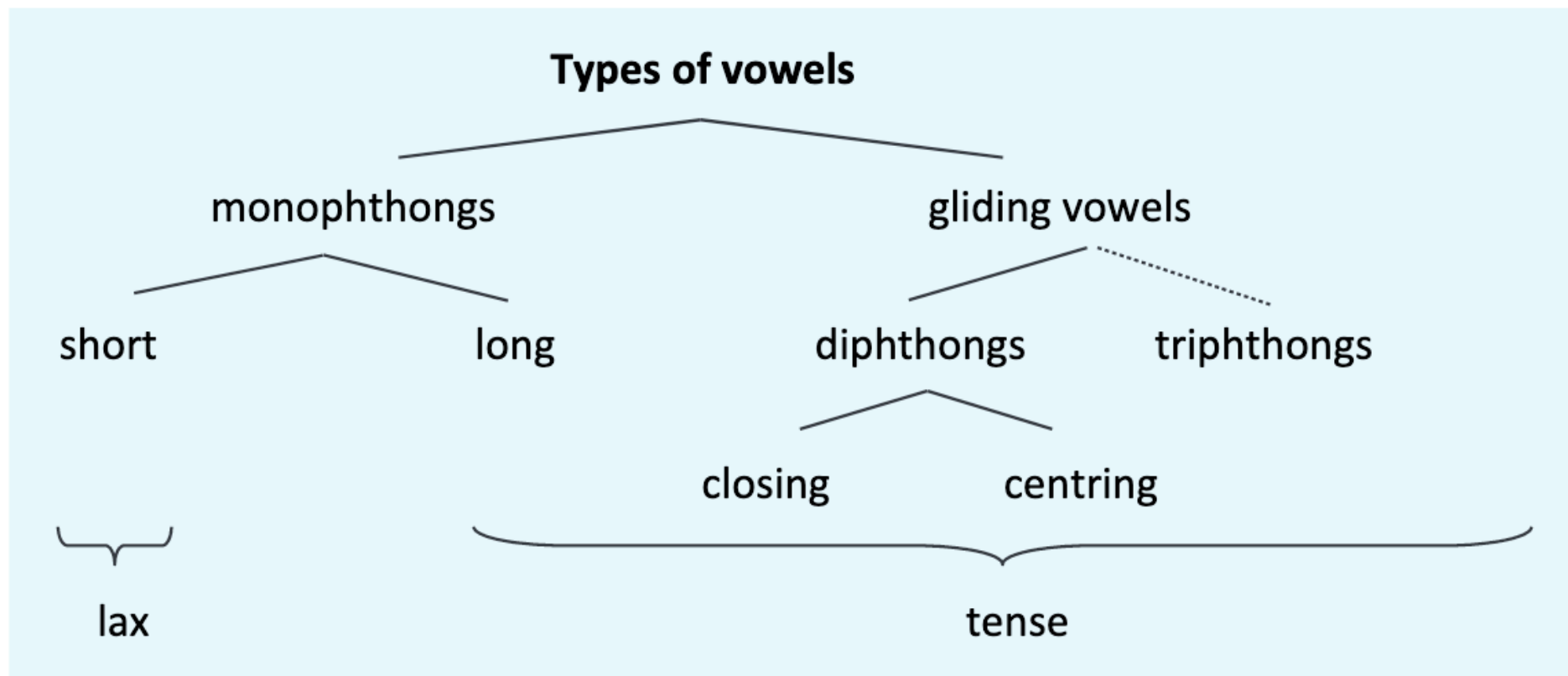


Figure 2.9: Closing diphthongs

Kortmann (2020): 36

Types of vowels



Kortmann (2020): 37

Exercise: identify vowel sounds

In the following sets of words, the sound of the vowel is the same in every case but one. Which of these words has a different vowel sound?

1. *sane, paid, eight, lace, mast*: *mast*: /æ/ instead of /eɪ/
2. *hoot, good, moon, grew, suit*: *good*: /ʊ/ instead of /u:/
3. *meat, steak, weak, theme, green*: *steak*: /eɪ/ instead of /i:/
4. *dud, died, mine, eye, guy*: *dud*: /ʌ/ instead of /aɪ/
5. *pen, said, death, mess, mean*: *mean*: /i:/ instead of /ɛ/
6. *ton, toast, both, note, toes*: *ton*: /ɒ/ instead of /əʊ/

Exercise: determine number of phonemes

How many phonemes are there in each of the following words?

Word	RP	GA	Phonemes
<i>physics</i>	<u>/ˈfɪzɪks/</u>	<u>←</u>	<u>6</u>
<i>axis</i>	<u>/ˈæksɪs/</u>	<u>←</u>	<u>5</u>
<i>laugh</i>	<u>/lɑːf/</u>	<u>/læf/</u>	<u>3</u>
<i>fishes</i>	<u>/ˈfɪʃɪz/</u>	<u>←</u>	<u>5</u>
<i>begged</i>	<u>/begd/</u>	<u>←</u>	<u>4</u>
<i>caught</i>	<u>/kɔːt/</u>	<u>/kɑːt/, /kɔːt/</u>	<u>3</u>
<i>batting</i>	<u>/ˈbætɪŋ/</u>	<u>←</u>	<u>5</u>
<i>knock</i>	<u>/nɒk/</u>	<u>/nɑːk/</u>	<u>3</u>
<i>quick</i>	<u>/kwɪk/</u>	<u>←</u>	<u>4</u>

Transcription

Transcription dictionaries

Jones, Daniel. 2011. Cambridge English pronouncing dictionary. Ed. By Peter Roach. 18th ed. Cambridge: Cambridge University Press.

<https://dictionary.cambridge.org/>

Alternative: Wells, John C. 2010. Longman pronunciation dictionary. 3rd ed. Harlow: Pearson Longman.

Exercise: evaluate transcriptions

Interactive version

Max challenge attempts exceeded. Please refresh the page to try again!

Reference version

Which of the two transcriptions (RP) is correct?

1. sixteen:

a. /sɪx'ti:n/

b. /sɪk'sti:n/ ✓

2. football:

a. /'fʊtbɔ:l/ ✓

b. /'fʊtbol/

3. strength:

a. /strenkθ/

b. /streŋθ/ ✓

4. umbrella:

a. /ʊm'brelə/

b. /ʌm'brelə/ ✓

5. hijacking:

a. /'haɪdʒækɪŋ/ ✓

b. /'haɪjækɪŋ/

6. handmade:

a. /,hand'meɪd/

b. /,hænd'meɪd/ ✓

7. chipping:

a. /'tʃɪppɪŋ/

b. /'tʃɪpɪŋ/ ✓

8. remain:

a. /rɪ'man/

b. /rɪ'meɪn/ ✓

Exercise: transcribe words

Transcribe the following words into phonemic IPA and indicate which variety your transcription is based on (Received Pronunciation (RP) or General American (GA)). Use stress marks.

Word	RP (UK)	GA (US)
1. <i>tea</i>	<u>/ti:/</u>	<u>←</u>
2. <i>cat</i>	<u>/kæt/</u>	<u>←</u>
3. <i>bar</i>	<u>/bɑ:/</u>	<u>/bɑ:r/</u>
4. <i>food</i>	<u>/fu:d/</u>	<u>←</u>
5. <i>foot</i>	<u>/fʊt/</u>	<u>←</u>
6. <i>pork</i>	<u>/pɔ:k/</u>	<u>/pɔ:rk/</u>
7. <i>maker</i>	<u>/ˈmeɪ.kə/</u>	<u>/ˈmeɪ.kə-/</u>
8. <i>phonology</i>	<u>/fəˈnɒl.ə.dʒi/</u>	<u>/fəˈnɑ:.lə.dʒi/</u>

Next week: Phonology

Read Kortmann ([2020](#)): ch. 2.2

References

Kortmann, Bernd. 2020. *English Linguistics: Essentials*. J.B. Metzler.